Environment 0

Data Context

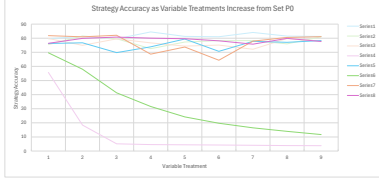
Each experiment has a new probability field generated.
This allows for a variety of strategy accuracy % values to be recorded as the probability field changes.
Therefore, each experimental treatment will be recorded as the average of 10 experiment % values.
This is in addition each experiment being an average of 10 trial % values over that experiment's unique probability field.
Overall, each % value listed will represent the average of 100 trial % values, for every 10 of which have a unique probability field.

Variable Minimum Values in Practice (Set P0)

N >= 10	Number of samples
T >= 10	Number of steps
S = 1	Step size
L >= 10	Number of trials (Stay as 10 as defined)
D >= 2	Number of dimensions
L = 1	Region length (Arbitrary scale factor)
A >= 100	Number of axis subdivisions
C >= 2	Number of cluster subdivisions
Rm = [0,1]	Range of mean values per axis
Rd = [0,1]	Range of standard deviation values per axis

*These values are justified in the notebook

Control %	74.454								
Experimental Treatment as Variation from Set P0									
N	15	20	25	30	35	40	45	50	55
%	79.652	81.28	79.555	84.477	81.307	81.04	84.093	81.497	81.228
T	20	30	40	50	60	70	80	90	100
%	75.469	81.596	79.63	72.691	77.308	78.31	78.282	76.017	81.01
S	2	3	4	5	6	7	8	9	10
%	79.804	74.948	79.583	76.709	74.423	75.158	72.178	79.788	77.951
D	4	6	8	10	12	14	16	18	20
%	55.747	18.422	5.149	4.576	4.416	4.293	4.1	3.9	3.8
A	200	300	400	500	600	700	800	900	1000
%	76.237	76.733	69.786	73.921	79.424	70.696	77.814	77.135	78.345
C	3	4	5	6	7	8	9	10	11
%	69.653	58.014	41.273	31.597	24.215	19.717	16.467	13.972	11.696
Rm	[0.25, 1.00]	[0.50, 1.00]	[0.75, 1.00]	[0.00, 0.75]	[0.00, 0.50]	[0.00, 0.25]	[0.20, 0.60]	[0.30, 0.70]	[0.40, 0.60]
%	81.851	80.996	82.11	68.712	75.774	64.404	77.977	80.559	81.111
Rd	[0.25, 1.00]	[0.50, 1.00]	[0.75, 1.00]	[0.00, 0.75]	[0.00, 0.50]	[0.00, 0.25]	[0.20, 0.60]	[0.30, 0.70]	[0.40, 0.60]
%	76.496	79.881	80.805	80.055	79.645	78.135	75.795	79.894	77.69
Treatment:	1	2	3	4	5	6	7	8	9



Environment 1

Data Context

Each experiment has a new probability field generated.
This allows for a variety of strategy accuracy % values to be recorded as the probability field changes.
Therefore, each experimental treatment will be recorded as the average of 5 experiment % values.
This is in addition each experiment being an average of 100 trial % values over that experiment's unique probability field.
Overall, each % value listed will represent the average of 500 trial % values, for every 100 of which have a unique probability field.

Variable Constant Values

X = 100	Number of trials, as defined
S = 1	Step size, T has the primary walk influence
L = 1	Region length, unit lattice grid
A = 100	Number of axis subdivisions, grid resolution
Rm = [0,1]	Range of mean values per axis
Rd = [0,1]	Range of standard deviation values per axis

Variable Minimum Values in Practice (Set P1)

T >= 10	Number of steps, mild distance from LHS
N >= 10	Number of samples, realistic
D >= 5	Number of dimensions, realistic
C >= 3	Number of cluster subdivisions, variety

*These values are justified in the notebook

Variable Minimum Values in Practice (Set P2)

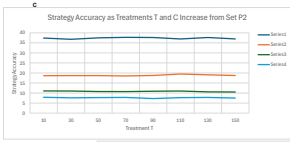
T >= 10	Number of steps, mild distance from LHS
N >= 10	Number of samples, realistic
D >= 3	Number of dimensions, reduce trial time
C >= 3	Number of cluster subdivisions, variety

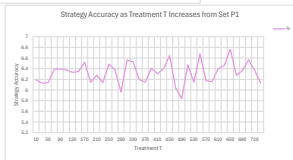
*These values are justified in the notebook

(Environment 1)

Experiment for T and C as Variation from Set P2

Treatment:	10	30	50	70	90	110	130	150	T
3	37.418	36.79	37.512	37.762	37.678	36.954	37.684	36.998	
4	18.678	18.7	18.702	18.522	18.764	19.492	19.076	18.738	
5	11.954	11.62	10.776	10.724	10.9	10.994	10.58	10.514	
6	7.896	7.592	7.74	7.835	7.26	7.702	7.82	7.526	
C									





(Environment 1)

Experiment for T as Variation from Set P1

Treatment:	T	%	Change
1	10	6.19	0
2	30	6.126	-0.064
3	50	6.14	0.014
4	70	6.39	0.25
5	90	6.398	0.008
6	110	6.376	-0.022
7	130	6.332	-0.044
8	150	6.346	0.014
9	170	6.524	0.178
10	190	6.144	-0.38
11	210	6.278	0.134
12	230	6.142	-0.136
13	250	6.484	0.342
14	270	6.38	-0.104
15	290	5.968	-0.412
16	310	6.562	0.594
17	330	6.526	-0.036
18	350	6.198	-0.328
19	370	6.149	-0.05
20	390	6.408	0.26
21	410	6.305	-0.103
22	430	6.4	0.095
23	450	6.646	0.246
24	470	6.034	-0.612
25	490	5.842	-0.192
26	510	6.47	0.628
27	530	6.146	-0.324
28	550	6.678	0.532
29	570	6.172	-0.506
30	590	6.156	-0.016
31	610	6.4	0.244
32	630	6.458	0.058
33	650	6.764	0.306
34	670	6.276	-0.488
35	690	6.364	0.088
36	710	6.568	0.204
37	730	6.37	-0.198
38	750	6.134	-0.236

Environment 2

Two detrimental errors for strategy accuracy calculation and for probability field generation have been resolved. This environment 2 will be first of the final conditions testing environments.

Data Context

Each experiment has a new probability field generated.
This allows for a variety of strategy accuracy % values to be recorded as the probability field changes.
Therefore, each experimental treatment will be recorded as the average of 5 experiment % values.
This is in addition each experiment being an average of 100 trial % values over that experiment's unique probability field.
Overall, each % value listed will represent the average of 500 trial % values, for every 100 of which have a unique probability field.

Control Set	
Constant Values	
X = 100	Number of trials, as defined
S = 1	Step size, T has the primary walk influence
L = 1	Region length, unit lattice grid
A = 100	subdivisions, grid resolution
Rm = [0,1]	Range of mean values per axis
Rd = [0,1]	an deviation values per axis
Variable Lower Bounds	
T >= 1	Number of steps, mild distance from LHS
N >= 10	Number of samples, realistic
D >= 3	Number of dimensions, reduce trial time
C >= 3	Number of cluster subdivisions, variety

Experiment for T as Variation from Set 3

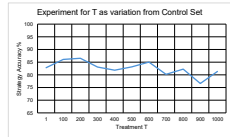
T	1	100	200	300	400	500	600	700	800	900	1000
%	82.806	86.094	86.532	83.88	81.842	83.15	85.026	80.188	82.256	76.584	81.348
Treatment:	1	2	3	4	5	6	7	8	9	10	11

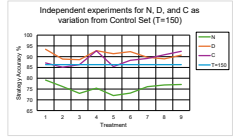
*average of 100 and 200 results

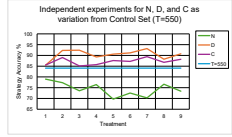
Control %	86.313	86.313	86.313	86.313	86.313	86.313	86.313	86.313	86.313
Experimental Treatment as Variation from Control Set for T=150									
N	20	30	40	50	60	70	80	90	100
%	79.164	76.166	73.002	75.418	71.976	73.118	76.106	76.884	77.104
D	4	5	6	7	8	9	10	11	12
%	93.44	88.902	88.592	92.8	91.39	92.336	89.72	89.012	90.72
C	4	5	6	7	8	9	10	11	12
%	87.116	85.126	86.346	92.728	85.424	88.314	89.396	90.888	92.492
Treatment:	1	2	3	4	5	6	7	8	9

* average of 500 and 600 results

Control %	84.088	84.088	84.088	84.088	84.088	84.088	84.588	84.088	84.088
Experimental Treatment as Variation from Set 3 for T=550									
N	20	30	40	50	60	70	80	90	100
%	78.984	77.322	73.556	76.448	69.698	72.518	70.26	76.598	73.38
D	4	5	6	7	8	9	10	11	12
%	85.486	92.304	92.398	89.27	90.364	91.164	93.17	88.226	90.754
C	4	5	6	7	8	9	10	11	12
%	85.496	89.024	85.098	85.69	87.61	87.206	89.418	88.78	88.208
Treatment:	1	2	3	4	5	6	7	8	9







Control Set

Constant Values	
Step size (S)	1
Region length per axis (L)	1
Subdivisions per axis (A)	100
Mean range (Rm)	[0, 1]
Standard deviation range (Rd)	[0.1, 1]
Trials (X)	100
Variable Lower Bounds	
Samples (N)	10
Dimensions (D)	3
Clusters per axis (C)	3
Steps (T)	1

Independent Experiments

Treatment	N	D	C
1	20	4	4
2	30	5	5
3	40	6	6
4	50	7	7
5	60	8	8
6	70	9	9
7	80	10	10
8	90	11	11
9	100	12	12